Mohith R

Bengaluru, India +91 63610 72033 mohithr4567@gmail.com linkedin.com/in/mohith-r-0763a72a4 github.com/Mohith-R17

Education

R V University, Bengaluru **Aug 2023 – May 2027** *B.Tech in Computer Science and Engineering*

CGPA: 8.06 / 10

Experience

Summer Internship in the centre of IOT and Edge Computing | GitHub

Bengaluru, India

- Designed and implemented an IoT-based air quality monitoring system for real-time environmental analysis.
- Integrated **ESP8266** with **DHT11** and **MQ135** sensors to collect temperature, humidity, and gas concentration data.
- Built real-time data visualization dashboards using the **Blynk IoT platform**.
- Implemented threshold-based alerts to notify users when pollution levels exceeded safe limits.
- Improved data reliability through sensor calibration and optimized data transmission logic.

Projects

Ephemetrix — OpenVR-Based IMU Motion Tracking System | GitHub **June 2025 – Present**

- Designed and implemented a low-cost, high-performance motion tracking prototype for the **OpenVR framework**, enabling real-time VR body orientation tracking.
- Engineered a tracking unit using an **ESP8266 NodeMCU** and **BNO055 9-DOF IMU**, establishing stable **I2C communication** for accurate pose estimation.
- Developed custom **C++ firmware (VRTracker**$_{s}ender.ino)tocapturefusedquaternionorientationandlinearaccelerationdataat$**100Hz**$.Implementeda$**low-latency**
- Validated end-to-end data integrity from hardware sensors to network packets, ensuring consistent virtual avatar pose updates.
- Designed the system architecture for future scalability to full-body tracking using **TCA9548A I2C multiplexing**.

SentientNPC — AI Voice-Interactive NPC Dialogue Framework | GitHub

- Architected an end-to-end, **voice-to-voice NPC interaction pipeline**, integrating STT, NLP, and TTS modules into a unified asynchronous framework for real-time response.
- Deployed a lightweight **Vosk Speech-to-Text engine**, optimized for edge computing to achieve **sub-200ms transcription latency** with partial result streaming for live feedback.
- Developed and trained a custom **Encoder–Decoder Transformer model** using **TensorFlow/Keras**, implementing multi-head self-attention and positional encoding for contextual dialogue generation.
- Integrated **Silero Text-to-Speech (Torch)** to synthesize expressive audio responses, utilizing **multi-threading** to prevent blocking during audio playback.
- Designed a robust **Tkinter-based GUI**, leveraging Python threading and queue mechanisms to manage concurrent audio streams, inference workloads, and UI updates under high load.

Technical Skills

Programming Languages: C, C++, Java, Python **Data Machine Learning**: Data Analysis, Machine Learning, Computer Vision, OpenCV, TensorFlow **Embedded IoT**: ESP8266, ESP32, RP2040, Embedded C, IoT Protocols **Web Technologies**: HTML, CSS, JavaScript

Achievements

NPTEL Design & Implementation of Human Computer Interfaces — Silver | Certificate